Light-Emitting Diode IV Characterization

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Introduction:

This experiment's main objective is to determine Planck's constant by observing the relationship between the applied voltages and forward current readings between light-emitting diodes (LEDs) of different wavelengths (colours). An LED acts as a p-n semiconductor junction, meaning that the electric current flowing through an LED can only travel in one direction. When the diode is placed in its forward-biased direction, electric conduction can only occur when the applied voltage overcomes the turn-on voltage. By using an Arduino-controlled circuit, the voltage applied to each LED can be varied at constant steps, providing accurate readings of each LED's turn-on voltages. From the semiconductor's band gap energy and its close relationship between the turn-on voltages and photon frequencies, a linearized relationship between the turn-on threshold voltages and photon frequencies can be derived, yielding an experimentally determined value for Planck's constant.

Theory:

A light emitting diode (LED) is a p-n junction diode in which, under forward bias, electrons from the n-type side recombine with holes on the p-type side, the electron drops from the conduction band into the empty valence band state. Because the electron ends up at a lower energy state, the extra energy is released, and in a direct band gap material like those in LEDs, it leaves as a photon. In indirect materials such as Si and Ge, recombination also requires a phonon, so most of the energy is lost as lattice vibrations and few photons are emitted. This is why LEDs are made from direct band gap III-V compounds as they are efficient at emitting light and alloy variations allow them to release different colours.

Forward bias decreases the potential barrier at the junction. As eV increases to the band gap energy E_g, the barrier is lowered enough that a large number of electrons from the n-type side and holes from the p-type side can cross the junction and become minority carriers. Because more carriers can cross as the barrier drops, the current increases exponentially with voltage. The minority carriers then recombine with majority carriers, emitting photons. The voltage at which the current begins to rise sharply is the turn-on voltage, V_t.

At a given temperature T, the current I given an applied voltage V is modeled by the Shockley equation:

$$I = I_s(e^{\frac{eV}{nk_BT}} - 1) \qquad (1)$$

where I_s is the reverse saturation current, e is the elementary charge, k_B is Boltzmann's constant and n is an ideality factor. Under forward bias with $V \gg \frac{k_BT}{e}$, the current rises exponentially. This is why the turn-on voltage is easiest to identify on a logarithmic current axis.

The emitted photon carries the band gap energy:

$$E_g = hv = \frac{hc}{\lambda} \qquad (2)$$

where v is the photon frequency, λ is its wavelength, h is Planck's constant and c is the speed of light. Conduction begins when the energy the applied voltage gives each electron reaches the band gap:

$$eV_t = E_g \qquad (3)$$

Substituting (2) into (3) gives:

$$V_t = (\frac{h}{e})v \qquad (4)$$

So V_t should be linear in v and pass through the origin, with slope $m = \frac{h}{e}$. With the accepted values $h = 6.626 \times 10^{-34}\, J{\cdot}s$ and $e = 1.602 \times 10^{-19}\, C$, the expected slope is $4.136 \times 10^{-15}\, V{\cdot}s$. In this experiment, $v = \frac{c}{\lambda}$ is found from each LED's listed wavelength and V_t from its I-V curve. Planck's constant then follows from the fitted slope as $h = m{\cdot}e$.

Procedure:

According to the diagram of the Arduino breadboard setup provided within the lab manual, the 3 jumper cables were connected between the DAC_VLED, INA219, Op-amp, and BJT. The code provided within the D2L homepage labelled LED_IV.txt was uploaded into the Arduino IDE software. A for loop was implemented within the code, changing the increment of i to i += 20, with a maximum of 4095, which gives voltage steps of about 0.025 V. In each iteration of the for loop, a delay of 10 ms was added before the voltage and current were read. The DAC output was then set to 0 V for a delay of 100 ms so that the LED could cool. A break condition was added within the for loop to ensure that the current through the LED does not exceed 100 mA. Once the code was compiled without any errors, the code was uploaded to the physical Arduino with a USB cable. One LED was placed onto the breadboard, ensuring that the longer side of the LED went into the Vin- pin of the INA219 and the shorter side was connected to the ground. The Serial Plotter tab was opened from the Arduino IDE, confirming that the graph "spiked" once the LED started blinking. Once the spikes were observed, the Serial Monitor tab was opened to obtain voltage and current values for each LED. One full LED blinking cycle was highlighted and pasted into Google Spreadsheets to create a graph between the voltage and current readings of one specific LED waveform, ensuring that an exponential curve was obtained to verify our breadboard setup and code. Once the graph was obtained, the LED was switched out for an LED of a different wavelength, and the voltage-current readings from the Serial Monitor were pasted into Google Spreadsheets once again to obtain a total of 7 graphs for each of the 7 different LEDs. The LEDs and jumper cables were then disconnected from the breadboard, and the Arduino IDE was then turned off.

Results and Calculations:

Table 1: LED Data of the Voltage and Current Observations Recorded by the Arduino Serial Recorder for Each Specific LED Colour/Wavelength: This table includes all voltage and current readings found for each LED colour, where the voltage in units of volts (V) is found in each leftmost

column, and the current in units of milliamperes (mA) is found in each rightmost column under each specific LED colour/wavelength.

Colour of LED (wavelength, λ)													
UV (395 nm)		**White** (450 nm)		**Blue** (461 nm)		**Green** (580 nm)		**Red** (658 nm)		**Yellow** (591 nm)		**IR** (940 nm)	
V (V)	**I (mA)**	**V (V)**	**I (mA)**	**V (V)**	**I (mA)**	**V (V)**	**I (mA)**	**V (V)**	**I (mA)**	**V (V)**	**I (mA)**	**V (V)**	**I (mA)**
1.02	-0.4	1.02	-0.4	1.02	-0.7	1.02	-0.7	1.02	-0.3	1.02	-0.5	0.77	-0.5
1.02	-0.6	1.02	-0.5	1.02	-0.3	1.02	-0.6	1.02	-0.5	1.02	-0.5	0.78	-0.4
1.02	-0.3	1.02	-0.3	1.02	-0.2	1.02	-0.3	1.02	-0.3	1.02	-0.3	0.77	-0.4
1.02	-0.5	1.02	-0.5	1.02	-0.4	1.02	-0.3	1.02	-0.4	1.02	-0.5	0.77	-0.4
1.02	-0.2	1.02	-0.2	1.02	-0.5	1.02	-0.4	1.01	-0.4	1.02	-0.2	0.78	-0.3
1.02	-0.3	1.02	-0.5	1.02	-0.5	1.02	-0.5	1.02	-0.6	1.02	-0.4	0.77	-0.2
1.02	-0.3	1.02	-0.7	1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.4	0.78	-0.7
1.02	-0.3	1.02	-0.2	1.02	-0.7	1.02	-0.7	1.02	-0.5	1.02	-0.3	0.78	-0.5
1.02	-0.2	1.02	-0.3	1.02	-0.4	1.02	-0.3	1.02	-0.5	1.02	-0.3	0.78	-0.5
1.02	-0.4	1.02	-0.4	1.02	-0.6	1.02	-0.2	1.02	-0.2	1.02	-0.4	0.78	-0.5
1.02	-0.6	1.02	-0.4	1.02	-0.7	1.02	-0.4	1.02	-0.5	1.02	-0.7	0.78	-0.4
1.02	-0.7	1.02	-0.7	1.02	-0.3	1.02	-0.5	1.02	-0.4	1.02	-0.6	0.78	-0.4
1.02	-0.1	1.02	-0.3	1.02	-0.2	1.02	-0.4	1.02	-0.4	1.02	-0.5	0.78	-0.5
1.02	-0.5	1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.3	1.02	-0.4	0.78	-0.7
1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.5	1.02	-0.4	1.02	-0.3	0.78	-0.6
1.02	-0.4	1.02	-0.7	1.02	-0.4	1.02	-0.5	1.02	-0.4	1.02	-0.4	0.78	-0.5
1.02	-0.6	1.02	-0.4	1.02	-0.3	1.02	-0.3	1.02	-0.5	1.02	-0.4	0.78	-0.5
1.02	-0.6	1.02	-0.5	1.02	-0.3	1.02	-0.2	1.02	-0.4	1.02	-0.1	0.77	-0.8
1.02	-0.4	1.02	-0.3	1.02	-0.5	1.02	-0.4	1.02	-0.4	1.02	-0.4	0.78	-0.5

1.02	-0.4	1.02	-0.3	1.02	-0.3	1.02	-0.4	1.02	-0.4	1.02	-0.5	0.78	-0.4
1.02	-0.5	1.02	-0.4	1.02	-0.5	1.02	-0.6	1.02	-0.5	1.02	-0.4	0.77	-0.5
1.02	-0.3	1.02	-0.1	1.02	-0.3	1.02	-0.5	1.02	-0.5	1.02	-0.2	0.78	-0.5
1.02	-0.2	1.02	-0.6	1.02	-0.3	1.02	-0.5	1.02	-0.2	1.02	-0.5	0.78	-0.6
1.02	-0.4	1.02	-0.5	1.02	-0.3	1.02	-0.4	1.02	-0.3	1.02	-0.5	0.78	-0.4
1.02	-0.7	1.02	-0.6	1.02	-0.5	1.02	-0.7	1.02	-0.5	1.02	-0.5	0.78	-0.6
1.02	-0.7	1.02	-0.2	1.02	-0.7	1.02	-0.5	1.02	-0.5	1.02	-0.7	0.77	-0.4
1.02	-0.5	1.02	-0.3	1.02	-0.4	1.02	-0.5	1.02	-0.7	1.02	-0.4	0.78	-0.4
1.02	-0.7	1.02	-0.1	1.02	-0.3	1.02	-0.4	1.02	-0.3	1.02	-0.6	0.78	-0.6
1.02	-0.4	1.02	-0.4	1.02	-0.4	1.02	-0.3	1.02	-0.5	1.02	-0.3	0.77	-0.3
1.02	-0.4	1.02	-0.4	1.02	-0.3	1.01	-0.7	1.02	-0.4	1.02	-0.4	0.78	-0.5
1.02	-0.3	1.02	-0.1	1.02	-0.4	1.02	-0.2	1.02	-0.5	1.02	-0.5	0.78	-0.5
1.02	-0.4	1.02	-0.6	1.01	-0.3	1.02	-0.3	1.02	-0.2	1.02	-0.5	0.78	-0.7
1.02	-0.2	1.02	-0.5	1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.3	0.78	-0.3
1.02	-0.6	1.02	-0.2	1.02	-0.7	1.02	-0.4	1.02	-0.6	1.02	-0.5	0.78	-0.5
1.02	-0.4	1.02	-0.3	1.02	-0.4	1.02	-0.4	1.02	-0.7	1.02	-0.5	0.78	-0.4
1.02	-0.5	1.02	-0.5	1.02	-0.6	1.02	-0.4	1.02	-0.7	1.02	-0.6	0.78	-0.8
1.02	-0.4	1.02	-0.6	1.02	-0.4	1.02	-0.8	1.02	-0.4	1.02	-0.4	0.78	-0.3
1.02	-0.4	1.02	-0.8	1.02	-0.3	1.02	-0.3	1.02	-0.5	1.02	-0.2	0.78	-0.5
1.02	-0.4	1.02	-0.4	1.02	-0.4	1.02	-0.3	1.02	-0.7	1.02	-0.4	0.78	-0.8
1.02	-0.6	1.02	-0.6	1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.4	0.78	-0.4
1.02	-0.7	1.02	-0.4	1.02	-0.3	1.02	-0.5	1.02	-0.5	1.02	-0.4	0.78	-0.4
1.02	-0.2	1.02	-0.4	1.02	-0.6	1.02	-0.4	1.02	-0.6	1.02	-0.5	0.78	-0.5
1.02	-0.6	1.02	-0.4	1.02	-0.4	1.02	-0.7	1.02	-0.3	1.02	-0.7	0.78	-0.5

1.02	-0.4	1.02	-0.4	1.02	-0.4	1.02	-0.4	1.02	-0.4	1.02	-0.4	0.78	-0.3
1.02	-0.5	1.02	-0.3	1.02	-0.6	1.02	-0.4	1.02	-0.4	1.02	-0.3	0.78	-0.3
1.02	-0.4	1.02	-0.5	1.02	-0.6	1.02	-0.8	1.01	-0.3	1.02	-0.3	0.77	-0.4
1.02	-0.5	1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.5	1.02	-0.3	0.78	-0.3
1.02	-0.4	1.02	-0.7	1.02	-0.4	1.02	-0.7	1.02	-0.7	1.02	-0.5	0.78	-0.3
1.02	-0.7	1.02	-0.6	1.02	-0.5	1.02	-0.3	1.02	-0.3	1.02	-0.5	0.78	-0.3
1.02	-0.3	1.02	-0.4	1.02	-0.4	1.02	-0.1	1.02	-0.2	1.02	-0.4	0.78	-0.4
1.02	-0.5	1.02	-0.5	1.02	-0.5	1.02	-0.3	1.02	-0.3	1.02	-0.4	0.78	-0.3
1.02	-0.3	1.02	-0.7	1.02	-0.5	1.02	-0.6	1.02	-0.6	1.02	-0.3	0.78	-0.7
1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.4	1.02	-0.4	1.02	-0.3	0.78	-0.4
1.02	-0.5	1.02	-0.4	1.02	-0.2	1.02	-0.4	1.02	-0.4	1.02	-0.3	0.78	-0.5
1.02	-0.5	1.02	-0.4	1.02	-0.4	1.02	-0.2	1.02	-0.5	1.02	-0.6	0.78	-0.6
1.02	-0.4	1.02	-0.7	1.02	-0.4	1.02	-0.5	1.02	-0.3	1.02	-0.4	0.79	-0.4
1.02	-0.1	1.02	-0.3	1.02	-0.2	1.02	-0.5	1.02	-0.4	1.02	-0.2	0.8	-0.4
1.02	-0.5	1.01	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.6	1.02	-0.3	0.81	-0.7
1.02	-0.4	1.02	-0.3	1.02	0	1.02	-0.4	1.02	-0.5	1.02	-0.4	0.82	-0.5
1.02	-0.4	1.02	-0.4	1.02	-0.5	1.02	-0.8	1.02	-0.5	1.02	-0.7	0.84	-0.4
1.02	-0.5	1.02	-0.4	1.02	-0.2	1.02	-0.6	1.02	-0.5	1.02	-0.5	0.85	-0.8
1.02	-0.1	1.02	-0.4	1.02	-0.7	1.02	-0.5	1.02	-0.2	1.02	-0.7	0.86	-0.7
1.02	-0.1	1.02	-0.1	1.02	-0.4	1.02	-0.5	1.02	-0.5	1.02	-0.4	0.88	-0.5
1.03	-0.6	1.03	-0.3	1.03	-0.2	1.03	-0.3	1.02	-0.3	1.03	-0.4	0.89	-0.3
1.04	-0.5	1.03	-0.2	1.03	-0.2	1.03	-0.4	1.04	-0.4	1.03	-0.5	0.91	-0.3
1.04	-0.4	1.04	-0.6	1.04	-0.4	1.04	-0.4	1.05	-0.5	1.05	-0.5	0.92	-0.4
1.06	-0.5	1.06	-0.6	1.06	-0.4	1.06	-0.3	1.06	-0.4	1.06	-0.5	0.94	-0.4

1.07	-0.5	1.07	-0.5	1.07	-0.6	1.07	-0.4	1.07	-0.5	1.07	-0.4	0.95	-0.2
1.09	-0.1	1.09	-0.3	1.09	-0.9	1.09	-0.2	1.09	-0.4	1.09	-0.6	0.96	-0.1
1.1	-0.3	1.1	-0.3	1.11	-0.3	1.1	-0.4	1.11	-0.4	1.11	-0.4	0.98	-0.3
1.13	-0.4	1.12	-0.7	1.13	-0.6	1.13	-0.4	1.13	-0.4	1.13	-0.2	1	0.2
1.15	-0.6	1.14	-0.4	1.15	-0.3	1.15	-0.7	1.15	-0.6	1.15	-0.6	1.01	0.3
1.16	-0.5	1.16	-0.5	1.16	-0.3	1.16	-0.3	1.16	-0.5	1.16	-0.5	1.02	0.6
1.18	-0.5	1.18	-0.3	1.18	-0.3	1.18	-0.4	1.19	-0.4	1.18	-0.8	1.04	1
1.2	-0.1	1.2	-0.3	1.2	-0.5	1.2	-0.7	1.2	-0.6	1.2	-0.5	1.05	1.5
1.23	-0.4	1.22	-0.3	1.22	-0.2	1.23	-0.6	1.23	-0.6	1.23	-0.6	1.06	2.4
1.25	-0.3	1.25	-0.4	1.25	-0.4	1.25	-0.5	1.25	-0.5	1.26	-0.7	1.08	3.2
1.28	-0.2	1.28	-0.8	1.28	-0.5	1.28	-0.3	1.27	-0.7	1.28	-0.5	1.1	4.7
1.3	-0.4	1.29	-0.6	1.3	-0.2	1.3	-0.3	1.3	-0.4	1.3	-0.5	1.1	6.2
1.32	-0.4	1.31	-0.4	1.31	-0.3	1.32	-0.2	1.31	-0.5	1.32	-0.3	1.12	7.5
1.34	-0.5	1.33	-0.3	1.34	-0.2	1.34	-0.4	1.34	-0.4	1.34	-0.4	1.13	9.2
1.36	-0.3	1.36	-0.1	1.36	-0.6	1.37	-0.5	1.36	0	1.36	-0.4	1.14	11.5
1.39	-0.6	1.38	-0.4	1.38	-0.4	1.39	-0.4	1.38	-0.2	1.39	-0.4	1.16	14
1.42	-0.4	1.41	-0.5	1.41	-0.5	1.42	-0.4	1.41	-0.4	1.42	-0.7	1.16	16.2
1.44	-0.4	1.43	-0.3	1.43	-0.5	1.43	-0.4	1.43	-0.2	1.43	-0.7	1.18	19
1.46	-0.5	1.46	-0.7	1.46	-0.3	1.45	-0.4	1.45	-0.4	1.45	-0.4	1.18	21.7
1.48	-0.4	1.47	-0.4	1.47	-0.4	1.47	-0.3	1.46	-0.5	1.48	-0.3	1.2	24.1
1.5	-0.5	1.5	-0.4	1.5	-0.4	1.5	-0.4	1.48	-0.6	1.5	-0.5	1.21	26.9
1.53	-0.5	1.52	-0.3	1.52	-0.5	1.52	-0.4	1.5	-0.5	1.52	-0.4	1.22	30.1
1.56	-0.2	1.55	-0.2	1.55	-0.2	1.55	-0.3	1.52	-0.3	1.55	-0.4	1.23	33.7
1.58	-0.4	1.57	-0.4	1.58	-0.5	1.57	-0.7	1.54	-0.5	1.57	-0.6	1.24	36.5

1.6	-0.3	1.59	-0.6	1.6	-0.5	1.59	-0.3	1.56	-0.3	1.59	-0.2	1.25	39.9
1.62	-0.5	1.62	-0.3	1.62	-0.4	1.61	-0.3	1.57	-0.4	1.6	-0.4	1.26	42.6
1.64	-0.2	1.64	-0.5	1.64	-0.5	1.63	-0.6	1.59	-0.3	1.62	-0.5	1.27	45.7
1.67	-0.5	1.67	-0.6	1.66	-0.5	1.65	-0.4	1.61	-0.4	1.64	-0.4	1.28	49.4
1.69	-0.3	1.69	-0.4	1.69	-0.3	1.68	-0.4	1.63	-0.3	1.66	-0.6	1.3	52.9
1.72	-0.4	1.72	-0.2	1.71	-0.5	1.7	-0.4	1.64	-0.2	1.68	-0.6	1.3	56.5
1.74	-0.4	1.74	-0.4	1.74	-0.7	1.71	-0.6	1.66	-0.2	1.7	-0.4	1.31	59.3
1.76	-0.4	1.76	-0.1	1.76	-0.5	1.72	-0.2	1.68	-0.2	1.71	-0.7	1.32	61.3
1.78	-0.4	1.78	-0.4	1.78	-0.5	1.74	-0.5	1.69	-0.1	1.73	-0.3	1.33	64.3
1.81	-0.4	1.8	-0.2	1.8	-0.4	1.75	-0.4	1.71	-0.3	1.74	-0.4	1.34	68.4
1.83	-0.5	1.83	-0.4	1.83	-0.5	1.77	-0.4	1.73	0.3	1.76	-0.3	1.35	72.5
1.86	-0.3	1.86	-0.5	1.86	-0.5	1.79	-0.4	1.74	0.1	1.78	-0.1	1.36	76.1
1.88	-0.5	1.88	-0.4	1.88	-0.3	1.8	-0.2	1.76	0.7	1.8	-0.1	1.37	79.3
1.91	-0.4	1.9	-0.4	1.9	-0.5	1.81	-0.1	1.77	0.9	1.82	-0.2	1.38	82.2
1.93	-0.7	1.92	-0.3	1.92	-0.6	1.83	0.1	1.79	1.3	1.83	0.1	1.39	85.1
1.95	-0.3	1.94	-0.4	1.94	-0.4	1.84	-0.3	1.8	1.9	1.84	0	1.4	88.4
1.98	-0.3	1.96	-0.5	1.97	-0.6	1.86	0.2	1.82	2.5	1.87	0.4	1.41	92.1
2	-0.5	2	-0.4	2	-0.3	1.88	0.3	1.84	3.3	1.89	0.8	1.42	95.8
2.03	-0.6	2.02	-0.6	2.02	-0.5	1.89	1.2	1.85	4.2	1.9	1	1.42	99.5
2.05	-0.4	2.04	-0.6	2.04	-0.3	1.9	1.6	1.86	5.4	1.91	1.2	1.44	102.8
2.07	-0.6	2.06	-0.5	2.06	-0.6	1.92	1.9	1.88	6.5	1.93	1.5		
2.09	-0.7	2.08	-0.4	2.08	-0.3	1.93	2.6	1.89	7.6	1.94	2		
2.12	-0.4	2.11	-0.4	2.11	-0.4	1.95	3.7	1.91	9.1	1.96	2.8		
2.14	-0.4	2.13	-0.3	2.14	-0.5	1.96	5.1	1.92	10.9	1.98	3		

2.17	-0.5	2.16	-0.4	2.16	-0.5	1.98	7	1.94	13.1	2	4		
2.19	-0.3	2.18	-0.5	2.18	-0.7	1.99	8.7	1.96	14.9	2.02	4.9		
2.21	-0.7	2.2	-0.3	2.2	-0.5	2	10.7	1.97	17.6	2.03	5.8		
2.24	-0.5	2.22	-0.3	2.22	-0.3	2.02	12.7	1.98	19.6	2.04	6.7		
2.25	-0.5	2.24	-0.3	2.25	-0.4	2.03	15.9	1.99	22.3	2.06	8.2		
2.28	-0.4	2.28	-0.4	2.28	-0.5	2.05	20.6	2	25	2.08	9.4		
2.3	-0.3	2.31	-0.4	2.3	-0.4	2.06	24.6	2.02	28.3	2.1	10.9		
2.33	-0.5	2.33	-0.6	2.32	-0.5	2.07	29.3	2.03	31.6	2.11	12.8		
2.36	-0.6	2.34	-0.5	2.34	-0.6	2.08	33.5	2.04	34	2.12	13.7		
2.38	-0.5	2.36	-0.5	2.36	-0.1	2.09	37.9	2.05	36	2.14	14.9		
2.4	-0.3	2.39	-0.3	2.38	-0.4	2.1	43	2.06	39.7	2.15	16.9		
2.42	-0.4	2.41	-0.6	2.39	-0.4	2.11	49.8	2.07	43.8	2.16	18.9		
2.45	-0.3	2.44	-0.4	2.41	-0.7	2.11	54.6	2.08	47.4	2.17	21.3		
2.48	-0.5	2.46	-0.4	2.43	-0.2	2.12	61	2.09	52	2.2	23.7		
2.5	-0.6	2.49	-0.3	2.45	-0.3	2.12	67.2	2.1	55.2	2.21	25.4		
2.52	-0.8	2.51	-0.5	2.46	-0.4	2.13	72.8	2.1	57.6	2.22	27.6		
2.54	-0.5	2.52	-0.5	2.47	-0.4	2.14	78.5	2.11	61	2.23	29.5		
2.56	-0.4	2.55	-0.2	2.49	-0.1	2.14	85.7	2.12	64.9	2.25	32		
2.59	-0.7	2.58	-0.7	2.5	-0.2	2.15	93	2.12	68.9	2.26	34.6		
2.62	-0.3	2.6	-0.4	2.52	-0.2	2.15	100.3	2.14	72.3	2.28	37.4		
2.64	-0.5	2.63	-0.4	2.54	0.4			2.14	75.7	2.29	39.9		
2.66	-0.5	2.65	-0.6	2.56	0.4			2.15	78.7	2.3	42.1		
2.67	-0.2	2.67	-0.4	2.56	0.5			2.15	81.2	2.31	44.7		
2.69	-0.2	2.69	-0.4	2.58	0.8			2.16	84.8	2.32	47.3		

2.72	-0.3	2.71	-0.5	2.6	1.3			2.16	88.9	2.34	50.5		
2.74	-0.1	2.73	-0.2	2.62	1.7			2.17	92	2.35	53.2		
2.76	-0.4	2.74	-0.4	2.63	1.9			2.17	94.9	2.36	56.2		
2.77	-0.4	2.76	-0.6	2.65	2.5			2.18	98.3	2.38	59		
2.78	-0.2	2.78	-0.3	2.67	3.1			2.18	101.2	2.38	61.4		
2.8	-0.3	2.8	-0.2	2.68	3.9					2.4	63.7		
2.82	-0.2	2.81	-0.3	2.7	4.4					2.41	67.1		
2.83	-0.4	2.83	-0.2	2.72	5.2					2.42	70.6		
2.85	0	2.84	-0.2	2.74	5.9					2.43	73.7		
2.86	0.1	2.86	-0.1	2.76	6.4					2.44	76.8		
2.88	0.3	2.87	-0.3	2.78	7					2.45	79.4		
2.89	0.4	2.88	0	2.79	7.9					2.46	81.7		
2.9	0.4	2.9	0.5	2.82	8.8					2.48	85.4		
2.92	0.7	2.91	1.1	2.83	9.7					2.48	88.6		
2.94	1.6	2.94	1.9	2.86	10.7					2.5	92.9		
2.95	2.3	2.95	2.6	2.87	11.5					2.51	96.4		
2.96	3.1	2.96	3.4	2.89	12.4					2.52	99.3		
2.98	3.8	2.97	4.6	2.9	13.2					2.53	102.2		
2.99	5	2.99	5.6	2.92	14.2								
3.01	6.3	3	7.2	2.94	15.3								
3.02	7.3	3.02	8.6	2.96	16.5								
3.04	8.9	3.04	10.4	2.98	18								
3.06	10	3.05	11.8	3	18.4								
3.07	11.1	3.06	13.3	3.02	19.9								

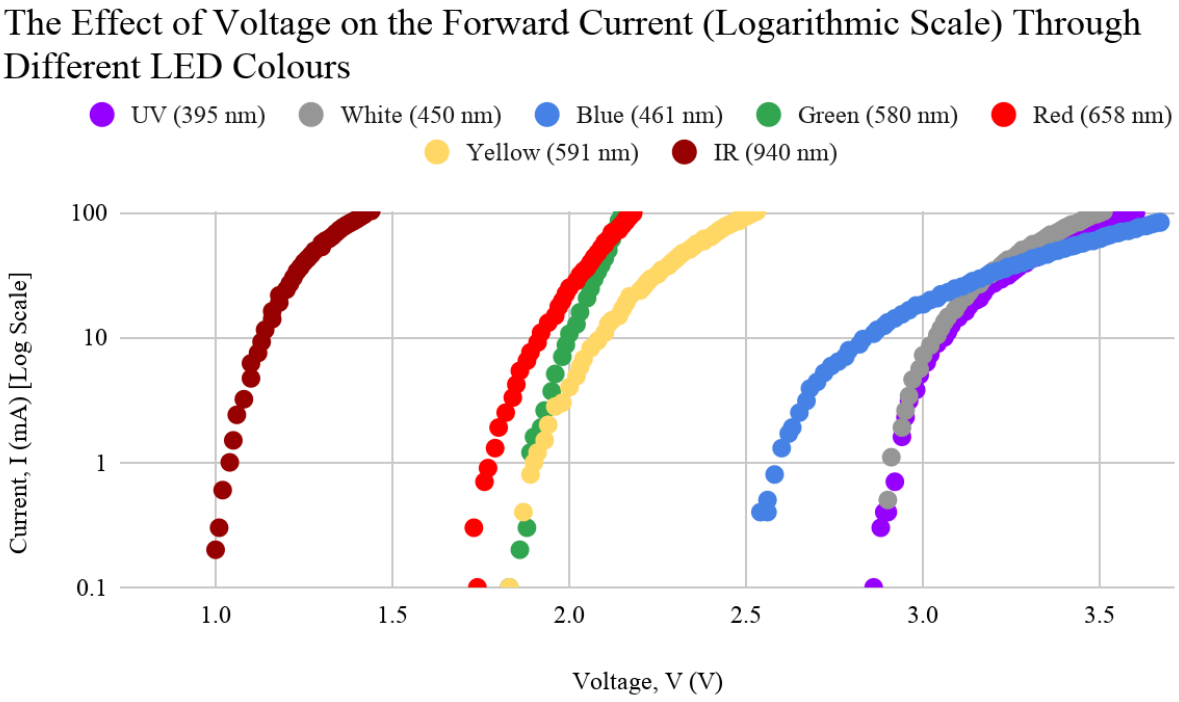


Figure 1: The Effect of Voltage on the Forward Current (Logarithmic Scale) Through Different LED Colours: The semi-logarithmic scale graph of the forward current as a function of the applied voltages for LEDs of different wavelengths can be used to identify the turn-on voltages (V_t).

Based on the readings observed in Figure 1, the turn-on voltages for each LED wavelength were measured when the forward current reading reached approximately 1 mA. From the log scale graph, the baseline sensor does not give an accurate reading at 0.1 mA, and not all of the LEDs provide a reading at the 0.1 mA current reading, proving that taking the turn-on voltages at that current would yield inaccurate results. Taking observations at the 1 mA current reading ensures that the diode has entered a stable state of exponential forward conduction, allowing us to obtain results with minimal uncertainty.

Table 2: The Observed Turn-on Threshold Voltages (V_t) with the Calculated Frequencies (ν) and Band Gap Energy Levels (E_g) of the different LED wavelengths: This table summarizes the data collected within the experiment to find the values of the turn-on voltages and band gap energies of each LED colour, where c is the speed of light ($3.0 \times 10^8\ m/s$) and e is the energy of an electron ($1.602 \times 10^{-19}\ C$).

LED Colour	**Wavelength, λ (nm)**	**Frequency, ν (Hz)** $\nu = c/\lambda$	**Turn-on Threshold Voltage, V_t (V)**	**Band Gap Energy, E_g (eV)** $E_g = eV_t$
Purple/UV	395	7.59×10^{14}	2.94 ± 0.025	2.94

White	450	6.67×10^{14}	2.91 ± 0.025	2.91
Blue	461	6.51×10^{14}	2.60 ± 0.025	2.60
Green	580	5.17×10^{14}	1.90 ± 0.025	1.90
Yellow	591	5.08×10^{14}	1.91 ± 0.025	1.91
Red	658	4.56×10^{14}	1.79 ± 0.025	1.79
IR (Infrared)	940	3.19×10^{14}	1.04 ± 0.025	1.04

The uncertainty value of $\pm$ 0.025V for the turn-on threshold voltages was determined from the voltage increments used within the Arduino IDE software. Since the DAC divides the 5V range into 4096 bits, the minimum voltage increment that can be set by the software is $\frac{5V}{4096\ bits} = 0.00122\ V/bit$. From the code, the for-loop increments the voltage by 20 bits each reading, which results in a voltage step increment of $0.025\ V/step$. Because the voltage is only incremented in factors of 0.025 Volts, the exact turn-on voltage could have occurred between two voltage increments, meaning that the uncertainty of the recorded values is approximately 0.025 Volts.

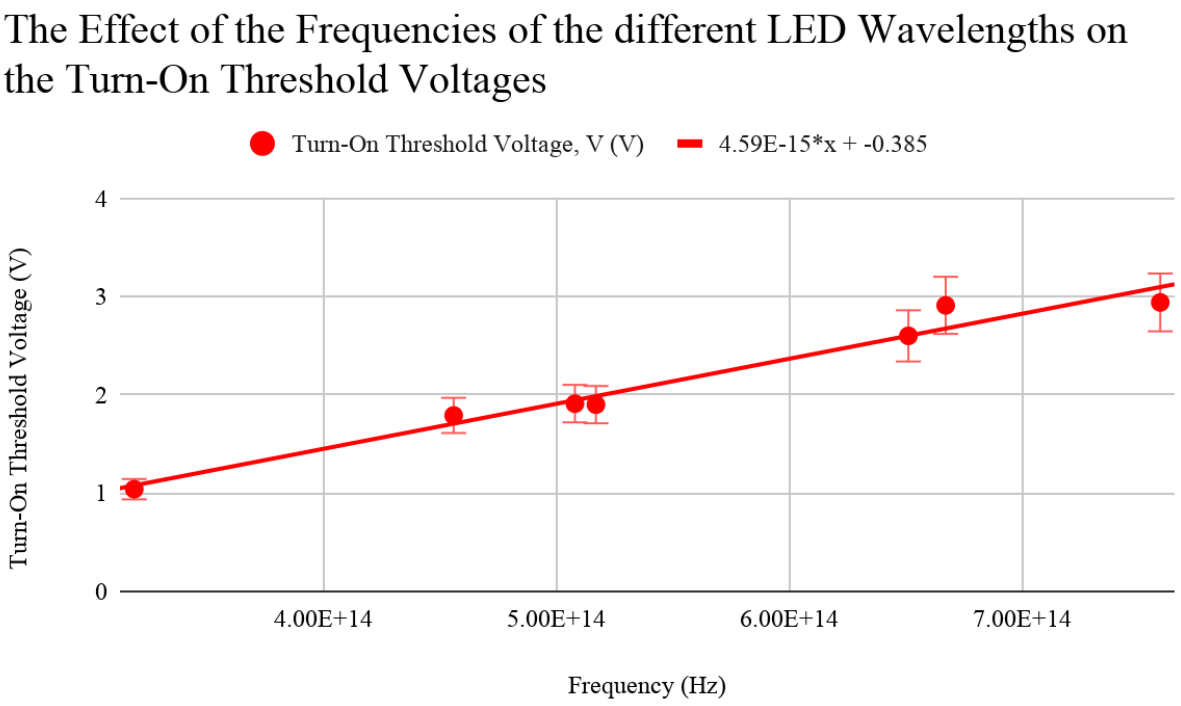


Figure 2: The Effect of the Frequencies of the different LED Wavelengths on the Turn-On Threshold Voltages: This graph shows the linear relationship between the turn-on voltage and the

frequencies of the LEDs, where the slope (m) is equal to Planck's constant over the energy of an electron, $m = \frac{h}{e}$.

To derive the linear relationship between the turn-on voltages and the frequency, equation (4) is used:

$$V_t = (\frac{h}{e})v$$

From Figure 2, the slope of this relationship is determined to be $m = 4.59 \times 10^{-15}$. Since $m = \frac{h}{e}$, Planck's constant can be calculated below:

$$m = \frac{h}{e} \Rightarrow h = m \cdot e \Rightarrow h = (4.59 \times 10^{-15})(1.602 \times 10^{-19})$$
$$h = 7.35 \times 10^{-34}\, J \cdot s$$

Therefore, the experimentally determined Planck's constant is $h = 7.35 \times 10^{-34}\, J \cdot s$.

$$\%\, Error = \frac{|Experimental\, Value - Theory\, Value|}{Theory\, Value} \times 100\% = \frac{|7.35\times10^{-34} - 6.626\times10^{-34}|}{6.626\times10^{-34}} \times 100\% = 10.7\%$$

When comparing the experimental and theoretical values for Planck's constant, it is observed that there is a discrepancy of approximately 10.7% between the two values. One reason for this discrepancy is that the white LED does not actually include a white energy band gap. The white LED is a high-energy blue or UV wavelength LED that includes a yellow layer of phosphor to simulate a broader light spectrum. This means that its turn-on voltage will be higher than what is theoretically expected since it is made up of a blue or UV LED. This is also observed within Figure 2, where the turn-on voltage of the white LED sits higher than the rest of the voltages relative to the linearized line of best fit. When removing the White LED data from Figure 2, the slope decreases to a value of $m = 4.32 \times 10^{-15}$ as observed in Figure 3 shown below. Determining the new Planck's constant, $h = (4.32 \times 10^{-15})(1.602 \times 10^{-19}) = 6.92 \times 10^{-34}\, J \cdot s$, yields a much more reasonable experimental error of only 4.2%.

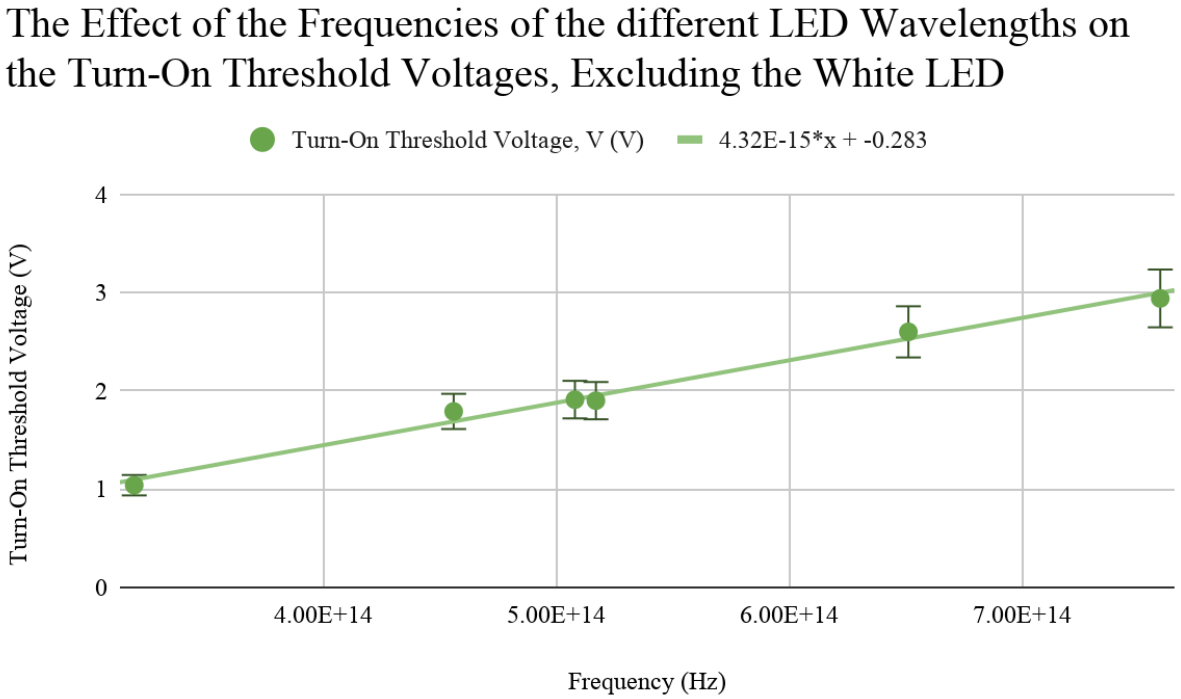


Figure 3: The Effect of the Frequencies of the Different LED Wavelengths on the Turn-On Threshold Voltages, Excluding the White LED: This graph shows the linear relationship between the turn-on voltage and the frequencies of the LEDs without including the data of the white LED, minimizing the experimental error within the lab.

Discussion and Conclusions:

Each LED's I–V curve showed the behaviour predicted by the Shockley equation: a flat baseline at low voltage while the LED is off, followed by a steep exponential rise once the LED turned on (Figure 1). In the baseline region the sensor reads between about −0.9 and 0 mA, which is its noise and offset. The turn-on voltage was therefore taken as the voltage at which the curve leaves this baseline and enters the exponential rise, read at a current of about 1 mA, the first level clearly above the noise. The same criterion was applied to all seven LEDs. The turn-on voltage increased as the wavelength decreased, from 1.04 V for the IR LED (940 nm) to 2.94 V for the UV LED (395 nm), as expected from $eV_t = \frac{hc}{\lambda}$. A plot of V_t against ν was approximately linear, with a slope of 4.59×10^{-15} V·s when all seven LEDs were included.

This slope gives h = 7.35×10^{-34} J·s, which is 10.9% above the accepted value of 6.626×10^{-34} J·s. The white LED accounts for much of this difference. It does not actually have a single band gap as it is a blue or UV emitter coated with a phosphor that converts part of the light to longer wavelengths. Its turn-on voltage is therefore set by the blue or UV chip, not by the listed 450 nm, so at 2.91 V it sits above the trend line in Figure 2. With the white LED removed, the slope drops to 4.32×10^{-15} V·s, giving h = 6.92 ×

10^{-34} J·s with an error of 4.4%. The white LED was excluded because its listed wavelength does not correspond to a true photon energy, not simply because it improves the fit, and both results are reported.

Both fitted lines also have a negative intercept (−0.385 V with all LEDs, −0.283 V without the white LED), whereas equation (4) predicts a line through the origin. Every LED except the white one has a measured V_t of about 0.1 to 0.3 V below $\frac{hc}{e\lambda}$ (1.04 V against 1.32 V for the IR LED and 2.94 V against 3.14 V for the UV LED). This is consistent with the exponential turn on because the current depends on $e^{(\frac{eV}{nk_BT})}$, a measurable 1 mA can flow at a voltage slightly below $\frac{Eg}{e}$. The relation $eV_t = E_g$ is therefore only approximate when V_t is read at a fixed current. Several other factors contribute to the remaining error. The ±0.025 V uncertainty from the DAC step size is small compared with the scatter of the points about the fitted line, which is up to about 0.2 V, so it is not the main limit but still does contribute to it. The differences between the LEDs matter more. The listed wavelengths are approximate, and an LED emits over a band of wavelengths, so ν is also uncertain. Heating was limited by pulsing the LED with a long off-time, however this does not fully eliminate it. The voltage was read by the INA219 at the LED itself, so voltage drops across the op-amp and transistor stages do not affect it.

In conclusion, the I–V curves of seven LEDs were measured with an Arduino-based curve tracer, and the turn-on voltage of each was found where the curve left its baseline, at about 1 mA on a logarithmic plot. The turn-on voltage increased linearly with photon frequency, as predicted by $V_t = (\frac{h}{e})\nu$. Including all LEDs, the slope gave h = 7.35 × 10^{-34} J·s (10.9% error). Excluding the white LED, which has no single band gap because of its phosphor coating, gave h = 6.92 × 10^{-34} J·s (4.4% error). The main sources of uncertainty were the turn-on criterion, the approximate wavelengths and the fact that $eV_t = E_g$ only holds approximately.

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Appendix:

1. The turn-on voltage is the forward voltage above which a diode carries a noticeable current. The ideal diode equation from Lab 3, $I = I_0[exp(\frac{V}{V_{th}}) - 1]$ with $V_{th} = \frac{k_B T}{e}$, shows that there is no sharp threshold. For $V \gg \frac{k_B T}{e}$ (about 0.026 V at room temperature) the current grows exponentially, increasing by a factor of e for every 26 mV of added voltage. Below the turn-on voltage the current is so small that it is negligible, so the diode appears to be off. Above it, the current climbs steeply, so the diode appears to switch on. The voltage at which this happens depends on the reverse saturation current, which is smaller for a larger band gap. Wider-gap semiconductors therefore need a larger voltage to reach the same current, which is why the turn-on voltage is related to E_g.

   Physically, the applied voltage lowers the potential barrier at the junction until many electrons from the n-type side and holes from the p-type side cross it and become minority carriers. In an ordinary diode these carriers recombine and release the energy as heat. In an LED the recombination happens across a direct band gap, so each event emits a photon. Turn-on is therefore seen as the steep rise in the I–V curve and as the LED begins to glow, dimly at first and then brighter as the current increases.

2. The two transistors differ in pin layouts. Viewed from the flat (labelled) face with the leads pointing down, the 2N3904 pins are in the order of emitter, base, collector from left to right while the TIP41C pins are in the order of base, collector, emitter. The wires connected to the 2N3904 would therefore have to move to different pins. The op-amp output would need to go to the TIP41C's left pin (base), the supply to the middle pin (collector) and the INA219 to the right pin (emitter). If the TIP41C were placed in the same three breadboard holes as the 2N3904, the op-amp output would need to connect to the middle pin, the supply would connect to the right pin and the INA219 would connect to the left pin for it to work.

3. Current through the LED rises exponentially with voltage while the circuit has no series resistor to limit it. A small increase in voltage near the top of the sweep can therefore cause a large increase in current. The break stops the sweep once the current exceeds 100 mA. If the Arduino were allowed to sweep the full 5 V or further, the current would exceed what the LED, the transistor and the current sensor are rated for. The LED would then overheat and burn out, which could also damage the other components.

4.

```
  // Signal path: Arduino → DAC → op-amp (buffer) → BJT → INA219 (measures V and
I) → LED → GND

#include <Wire.h>              // I2C library, used to talk to the DAC and INA219
#include <SPI.h>               // SPI library (from the starter code, not used)
#include <Adafruit_MCP4725.h>  // DAC library (voltage source)
#include <Adafruit_INA219.h>   // INA219 current sensor library

Adafruit_INA219 ina219;         // INA219 sensor object (reads LED voltage and current)
Adafruit_MCP4725 dac_LEDsupply; // DAC object (sets the voltage supplied to the LED)
#define DAC_RESOLUTION (9)      // From the starter code, not used

float voltageAcrossLED;         // LED voltage reading (V)
float currentAcrossLED;         // LED current reading (mA)

void setup(void) {             // Runs once at startup
    Serial.begin(9600);        // Starts serial communication with the computer
    ina219.begin();            // Starts the INA219 current sensor
    dac_LEDsupply.begin(0x62); // Starts the DAC at its I2C address
}

void loop() {                  // Repeats, so a new sweep starts each time
    // Each step of i += 20 changes the DAC output by 20 x (5 V / 4096) = 0.0244 V
    for (int i = 0; i ≤ 4095; i += 20) {   // Steps the DAC code from 0 to 4095
        dac_LEDsupply.setVoltage(i, false); // Sets the DAC voltage, which goes through
the op-amp and BJT to the LED
        delay(10);                          // LED on for 10 ms
        voltageAcrossLED = ina219.getBusVoltage_V(); // Reads the LED voltage from the
INA219
        Serial.print(voltageAcrossLED);     // Sends the voltage to the computer
        Serial.print(" ");                  // Space between the voltage and current

        currentAcrossLED = ina219.getCurrent_mA(); // Reads the LED current from the
INA219
        Serial.print(currentAcrossLED);     // Sends the current to the computer

        if (currentAcrossLED > 100.0) {     // Checks if the current is above 100 mA
            i = 4096;                       // Ends the sweep to protect the LED and
circuit
        }

        dac_LEDsupply.setVoltage(0, false); // Turns the LED supply off
        delay(100);                         // LED off for 100 ms to cool
        Serial.println("");                 // Starts a new line for the next reading
```

```
        }
    }
```